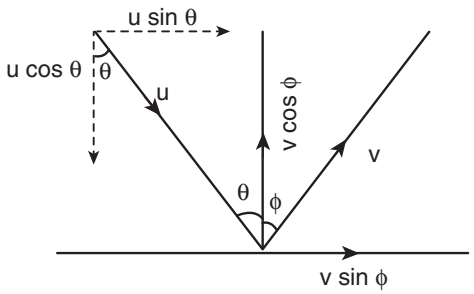
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Solution:

Applying component of velocities,



The x - component of velocity is

$$u\sin\theta = v\sin\phi \qquad (1)$$

The magnitude of y – component of velocity is not same, therefore, using coefficient of restitution,

$$e = \frac{v\cos\varphi}{u\cos\theta} \qquad (2)$$

Squaring (1) and (2) and adding we get

$$v^2\sin^2\varphi = u^2\sin^2\theta$$
$$v^2\cos^2\varphi = e^2u^2\cos^2\theta$$
$$\text{adding}$$
$$v^2 = u^2\sin^2\theta + e^2u^2\cos^2\theta$$
$$\therefore v^2 = u^2\left[\sin^2\theta + e^2\cos^2\theta\right]$$
$$v = u\sqrt{\sin^2\theta + e^2\cos^2\theta}$$

8. A particle of mass m is fixed to one end of a light spring of force constant k and un-stretched length l. It is rotated with an angular velocity ω in horizontal circle. What will be the length increase in the spring?

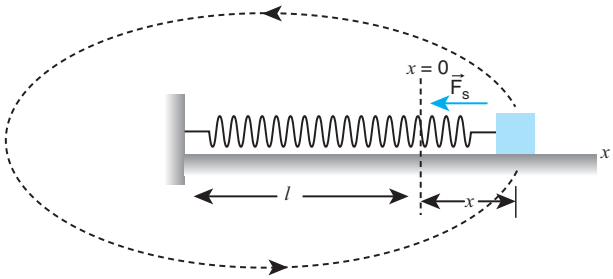
Solution:

Mass spring = m

Force constant = k

Un-stretched length = l

Angular velocity = ω



Let 'x' be the increase in the length of the spring.

The new length = (l+x) = r

When the spring is rotated in a horizontal circle,

Spring force = centripetal force.

$$kx = m\omega^2(l+x)$$
$$x = \frac{m\omega^2 l}{k - m\omega^2}$$

9. A gun fires 8 bullets per second into a target X. If the mass of each bullet is 3 g and its speed 600 ms^{-1}. Then, calculate the power delivered by the bullets.

Solution:

Power = work done per second = total kinetic energy of 8 bullets per second

$$P = 8\times(\text{kinetic energy of each bullet per second})$$
$$= 8\times\frac{1}{2}\times\left(3\times10^{-3}\right)\times\left(600\right)^2$$

P = 4320W

P = 4.320 kW

SOLVED EXAMPLE UNIT-5

1. Three particles of masses m_1 = 1 kg, m_2 = 2 kg and m_3 = 3 kg are placed at the corners of an equilateral triangle of side